

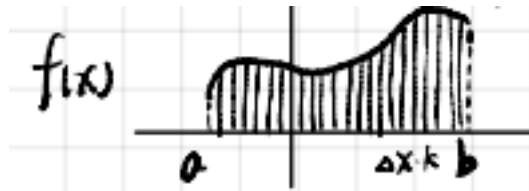
Chapter 3

Complex Integral

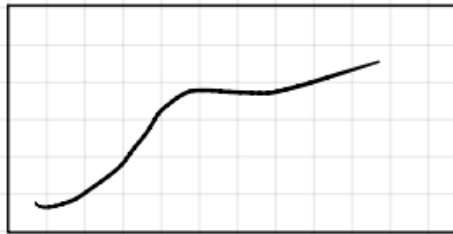
1. Line Integral

In the real domain, given a function $f(x)$, its integral is defined by

$$\int_a^b f(x) dx = \sum_{k=k_a}^{k_b} f(k\Delta x) \cdot \Delta x$$

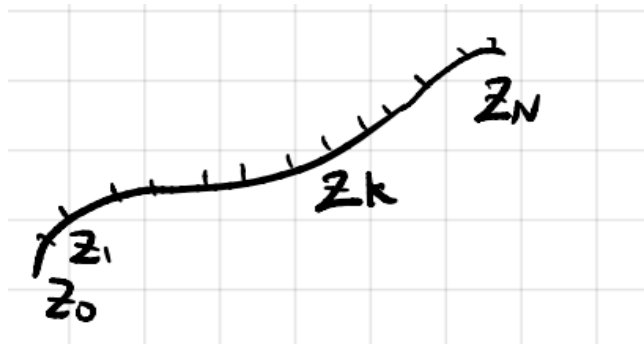


A line integral is integral in which the function to be integrated is determined along a curve in the coordinate system. The function which is to be integrated may be either a scalar field or a vector field. We can integrate a scalar-valued function or vector-valued function along a curve.



In the complex domain, we can divide the curve into small segments and define the line integral for a complex function $f(z)$

$$\int_L f(z) dz = \sum_{k=1}^M f(z_k) \cdot \Delta z_k$$



Example 1

Given a function $f(z) = z^2$, compute its line integral along the following path:



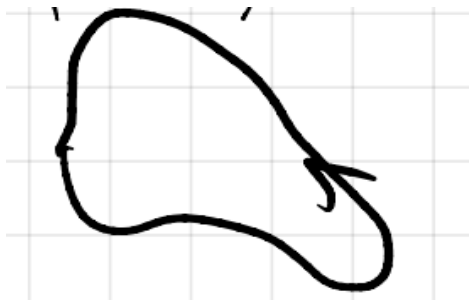
This path has two segments L_1 and L_2 . First, we find the parametric representation for each segment:

$$L_1: z = t, \quad 0 \leq t \leq 1$$

$$L_2: z = 1 + i \cdot t, \quad 0 \leq t \leq 1$$

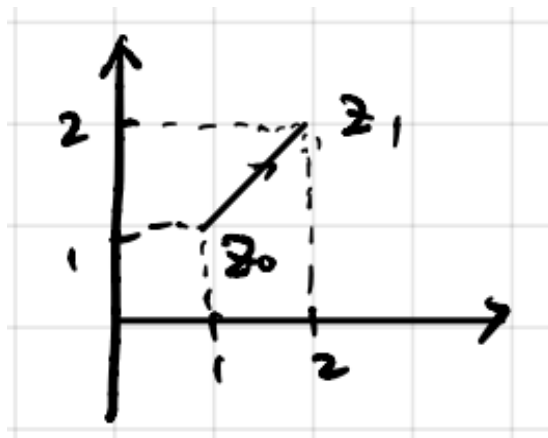
$$\begin{aligned} \int_L f(z) dz &= \int_{L_1} f(z) dz + \int_{L_2} f(z) dz \\ &= \int_0^1 x^2 dx + \int_0^1 (1 + it)^2 i dt \\ &= \frac{1}{3} x^3 \Big|_0^1 + i - i \frac{1}{3} t^3 \Big|_0^1 - 2 \frac{1}{2} t^2 \Big|_0^1 \\ &= \frac{2}{3} i + \frac{2}{3} \end{aligned}$$

Contour Integral is computed counter-clockwise on the contour.



Example 2

Find the line integral $\int_L z^2 dz$ for the following path



Step 1: find the parametric representation of the path

$$z = t + it \quad 1 \leq t \leq 2$$

$$dz = dt + i dt$$

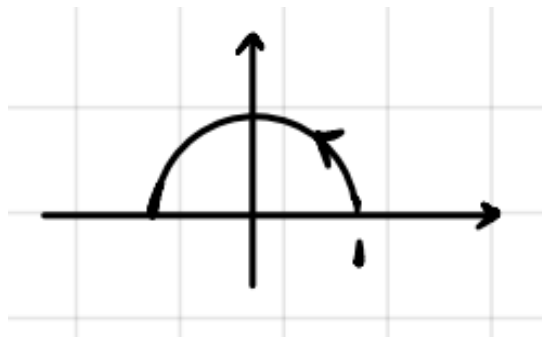
$$\frac{dz}{dt} = 1 + i \Rightarrow dz = (1 + i) dt$$

Step 2: compute the integral

$$\begin{aligned}
 \int_2 z^2 dz &= \int_1^2 (t + it)^2 (1 + i) dt \\
 &= (1 + i)^3 \int_1^2 t^2 dt \\
 &= (1 + i)^3 \cdot \frac{1}{3} t^3 \Big|_1^2 \\
 &= \frac{7}{3} (1 + i)^3
 \end{aligned}$$

Example 3

Find the line integral $\int_L z^2 dz$ on the following path



We first find the parametric representation of the path:

$$z = e^{i\theta} (0 \leq \theta \leq \pi)$$

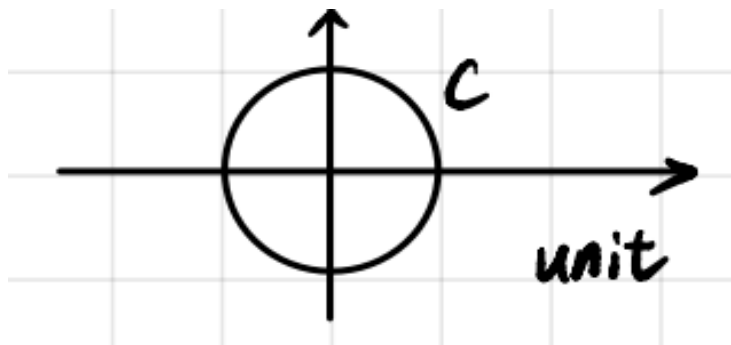
Then, we have

$$\begin{aligned}
 \frac{dz}{d\theta} &= ie^{i\theta} \\
 dz &= ie^{i\theta} d\theta
 \end{aligned}$$

Now, the integral becomes

$$\begin{aligned}
 \int_L z^2 dz &= \int_0^\pi e^{i2\theta} \cdot ie^{i\theta} d\theta \\
 &= i \int_0^\pi e^{3i\theta} d\theta \\
 &= i \cdot \frac{1}{3i} e^{3i\theta} \Big|_0^\pi = -\frac{2}{3}
 \end{aligned}$$

Example 4



$$z = e^{i\theta} \quad 0 \leq \theta \leq 2\pi$$

$$\frac{dz}{d\theta} = ie^{i\theta} \Rightarrow dz = ie^{i\theta} d\theta$$

$$\oint_C z^n dz = \int_0^{2\pi} e^{in\theta} \cdot ie^{i\theta} d\theta$$

$$= i \int_0^{2\pi} e^{i(n+1)\theta} d\theta$$

$$\text{If } n = -1 \quad \oint_C z^n dz = i \cdot 2\pi$$

$$\text{If } n \neq -1 \quad \oint_C z^n dz = i \frac{1}{i(n+1)} \cdot \int_0^{2\pi} e^{i(n+1)\theta} d\theta = 0$$

We conclude that, on the unit circle C ,

$$\oint_C z^n dz = \begin{cases} 2\pi i & n = -1 \\ 0 & n \neq -1 \end{cases}$$

Exercise

Find the line integral $\int_L z^2 dz$ along the following path



$$L = L_1 \oplus L_2$$

$$\int_L f(z) dz = \int_{L_1} f(z) dz + \int_{L_2} f(z) dz$$

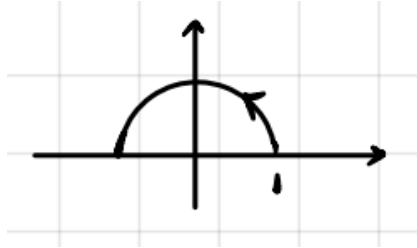
The complex function can take the following forms, $z = x + iy$

$$f(z) = \operatorname{Re}(z) = x$$

$$f(z) = \operatorname{Im}(z) = y$$

Example 5

Computer the line integral $\int_L \operatorname{Re}(z) dz$ on the following path



We can use the Euler formula

$$\begin{aligned} z &= e^{i\theta} \\ dz &= ie^{i\theta} d\theta \\ \operatorname{Re} z &= \frac{e^{i\theta} + e^{-i\theta}}{2} \\ &= \int_0^\pi \frac{e^{i\theta} + e^{-i\theta}}{2} ie^{i\theta} d\theta \\ &= \frac{i}{2} \int_0^\pi (e^{i2\theta} + 1) d\theta = \frac{\pi i}{2} \end{aligned}$$

2. Review of Green Formula

Green Formula

$$\oint_c P(x, y) dx + Q(x, y) dy = \iint_s \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

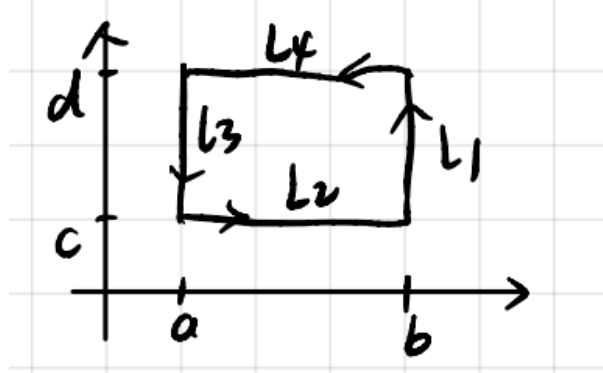
$$\iint f(x, y) dx dy = \sum_k \sum_M f(x_k, y_m) \cdot \Delta x \Delta y$$

Proof of the Green Formula:

In the one-dimensional case,

$$\int_a^b f'(x)dx = f(b) - f(a)$$

Now, let us consider a simple 2-D region, a rectangle



First, we have

$$\begin{aligned} & \iint_S \left(\frac{\partial Q(x, y)}{\partial x} - \frac{\partial P(x, y)}{\partial y} \right) dx dy \\ &= \left(\int_c^d Q(b, y) dy - Q(a, y) dy \right) - \int_a^b (P(x, d) - P(x, c)) dx \\ & \oint_C P dx + Q dy \end{aligned}$$

Also,

$$\oint_C P dx = \int_{L_1} P dx + \int_{L_2} P dx + \int_{L_3} P dx + \int_{L_4} P dx$$

On L_1 and L_3 , $dx = 0$, we have:

$$\oint_C P dx = 0 + \int_a^b P(x, c) dx + 0 - \int_a^b P(x, d) dx$$

Thus:

$$\oint_C P dx = \int_a^b (P(x, c) - P(x, d)) dx$$

For the Q function,

$$\oint_C Q dy = \int_{L_1} Q dy + \int_{L_2} Q dy + \int_{L_3} Q dy + \int_{L_4} Q dy$$

We have

$$\oint_C Q dy = \int_c^d Q(b, y) dy + 0 - \int_c^d Q(a, y) dy + 0$$

Thus:

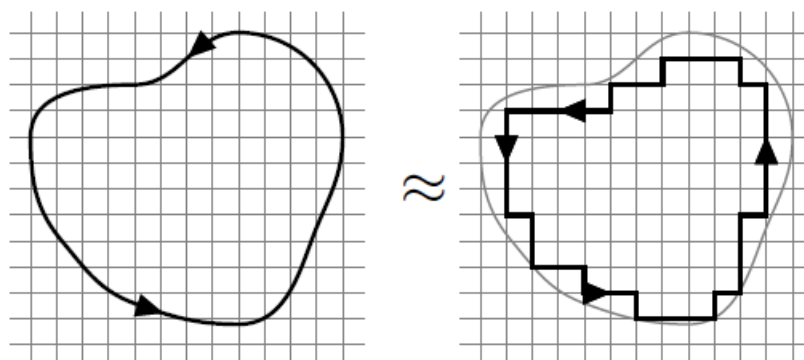
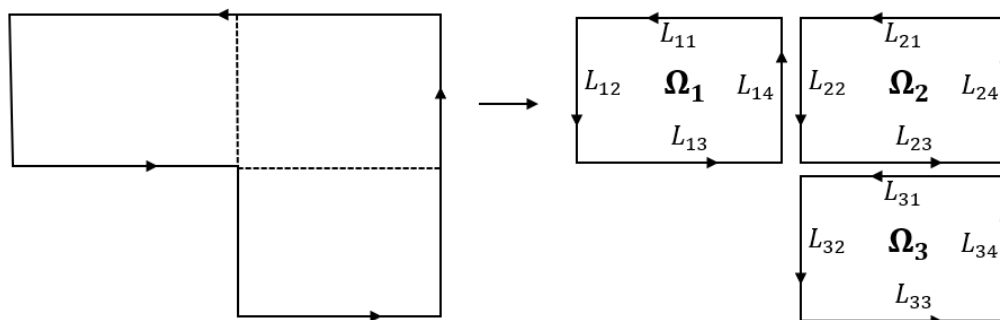
$$\oint_C Q dy = \int_c^d (Q(b, y) - Q(a, y)) dy$$

Finally, we have

$$\oint_{C_0} P dx + Q dy = \iint_{\Omega_0} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy, \quad \Omega_0 \text{ is rectangle}$$

Which proves the Green formula on the rectangle.

For line integrals, when adding two rectangles with a common edge the common edges are traversed in opposite directions so the sum is just the line integral over the outside boundary. Similarly when adding a lot of rectangles: everything cancels except the outside boundary. This extends Green's Theorem on a rectangle to Green's Theorem on a sum of rectangles. Since any region can be approximated as closely as we want by a sum of rectangles, Green's Theorem must hold on arbitrary regions.



3. Cauchy Theorem

Given a Complex function $f(z)$, the contour integral can be written as

$$\begin{aligned} z &= x + iy \\ f(z) &= u(x, y) + iv(x, y) \\ dz &= dx + idy \end{aligned}$$

$$\begin{aligned} \Rightarrow \oint_C f(z) dz &= \oint_C (u + iv)(dx + idy) \\ &= \oint_C u dx - v dy + i \oint_C v dx + u dy \end{aligned}$$

$$\begin{aligned} \oint_C f(z) dz &= \oint_C u dx - v dy + i \oint_C v dx + u dy \\ \oint_C P dx + Q dy &= \int_{\Omega} Q_x - P_y dxdy \\ \Rightarrow \oint_C u dx - v dy &= \int_{\Omega} -v_x - u_y dxdy \\ \oint_C v dx + u dy &= \int_{\Omega} u_x - v_y dxdy \end{aligned}$$

$f(z)$ 是可解析的, 那么根据 Cauchy-Riemann condition, 可得,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

把式子代入上面两个式子中, 得到

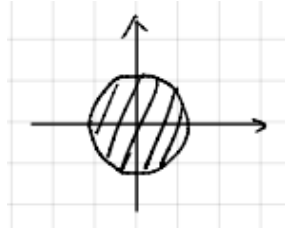
$$\begin{aligned} \oint_C u dx - v dy &= \int_{\Omega} -v_x - u_y dxdy \\ \oint_C v dx + u dy &= \int_{\Omega} u_x - v_y dxdy \\ \Rightarrow \oint_C f(z) dz &= 0 \end{aligned}$$

所以我们可以得出: $f(z)$ 是可解析的 $\Rightarrow$ **Cauchy-Riemann condition**

$$\Rightarrow \text{Green formula } \oint_C f(z) dz = 0$$

使用 Green Formula

Example 1: $f(z) = z^n, n > 0$. 求 $\oint_{|z|=1} z^n dz$, z 如图所示



$$f'(z) = n \cdot z^{n-1}$$

$$\oint_{|z|=1} z^n dz = 0$$

Example 2: Find the $\oint_{|z|=1} \sin(z^2 + 4z) dz$

Since $h(z) = \sin(z)$ and $g(z) = z^2 + 4z$ are both analytic, then according to the Chain rule, their compound function $f(z) = h(g(z)) = \sin(z^2 + 4z)$ is also analytic.

According to the Cauchy theorem, we have

$$\oint_{|z|=1} \sin(z^2 + 4z) dz = 0$$

Example 6

Evaluate the following integral $\int_0^{2\pi} \sin^{2n} \theta d\theta$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} = \frac{z - z^{-1}}{2i}$$

$$dz = i d\theta \cdot e^{i\theta} \Rightarrow d\theta = \frac{1}{ie^{i\theta}} dz = \frac{1}{iz} dz$$

$$\int_0^{2\pi} \sin^{2n} \theta d\theta = \int_{|z|=1} \left(\frac{z - z^{-1}}{2i} \right)^{2n} \frac{1}{iz} dz$$

Note that

$$(x + y)^n = \sum_{k=0}^n \frac{n!}{k! (n-k)!} x^k y^{n-k}$$

$$\begin{aligned}\int_0^{2\pi} \sin^{2n} \theta d\theta &= \oint_{|z|=1} \left(\frac{z^2 - 1}{2iz} \right)^{2n} \frac{dz}{iz} \\ &= \oint_{|z|=1} \frac{1}{(2i)^{2n}} \cdot \frac{1}{z^{2n+1}} \cdot \frac{1}{i} \cdot \sum_{l=0}^{2n} a_l z^{2l} (-1)^l dz, \quad a_l = \frac{n!}{l!(n-l)!}\end{aligned}$$

Remember that

$$\oint_c z^n dz = \begin{cases} 2\pi i & n = -1 \\ 0 & n \neq -1 \end{cases}$$

When we compute the integral of the summation, only the z^{-1} term will not be zero. Therefore, we have

$$\int_0^{2\pi} \sin^{2n} \theta d\theta = \pi \cdot \frac{1}{2^{2n-1}} \cdot \frac{(2n)!}{n! n!}$$

Given a polynomial with real coefficients:

$$P_n(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

where $a_n \cdots a_0$ are real, Then, $P_n(z)$ must have at least one root z_0 , such that $P_n(z) = 0$

$P(z_0) = 0$ $P(x) = (x - z_0)Q(x)$ Again, $Q(x)$ will have a root z_1 $\therefore P(x) = (x - z_0)(x - z_1)Q_1(x)$ Finally, we have $P(x) = (x - z_0)(x - z_1)(x - z_2) \cdots (x - z_{n-1}) \cdot C$ Conclusion, a polynomial of order n has n roots.
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Proof

$$P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

$P(z)$ is a polynomial of order n , a_k is real, z is complex, $a_n \neq 0$

We prove this by contradiction. Suppose that $P(z) \neq 0$ for any z

$P(z)$ is continuous, if z is real, then $P(z)$ must be all positive or all negative

$$p(z) \neq 0, \quad \text{let } z = 2 \cos \theta, 0 \leq \theta \leq 2\pi \Rightarrow p(2 \cos \theta) \neq 0.$$

$$\begin{aligned} \int_0^{2\pi} \frac{d\theta}{p(2\cos\theta)} &\neq 0 \\ \Rightarrow z = e^{i\theta} \quad \cos\theta &= \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z + z^{-1}}{2} \\ dz &= e^{i\theta} \cdot i \cdot d\theta \Rightarrow d\theta = \frac{1}{iz} dz \\ \therefore \int_0^{2\pi} \frac{d\theta}{p(2\cos\theta)} &= \oint_{|z|=1} \frac{1}{p(z + z^{-1})} \cdot \frac{1}{iz} dz \end{aligned}$$

If we can prove that $\frac{1}{p(z+z^{-1})} \cdot \frac{1}{iz}$ is analytic, then according to Cauchy theorem, we have

$$\oint_{|z|=1} \frac{1}{p(z + z^{-1})} \cdot \frac{1}{iz} = 0$$

So, the next step is to prove that $\frac{1}{p(z+z^{-1})} \cdot \frac{1}{iz}$ is analytic

$$\begin{aligned} P(z) &= a_n z^n + \dots + a_1 z + a_0 \\ P(z + z^{-1}) &= (z^{-1})^n [Q_{2n}(z)], \\ \oint_{|z|=1} \frac{1}{P(z + z^{-1})} \cdot \frac{1}{iz} &= \frac{1}{i} \oint_{|z|=1} \frac{1}{Q_{2n}(z)} \cdot \frac{1}{z^{-n} \cdot z} dz \\ &= \frac{1}{i} \oint_{|z|=1} \frac{z^{n-1}}{Q_{2n}(z)} dz \\ z \neq 0 : P(z) \neq 0 &\Rightarrow P(z + z^{-1}) \neq 0 \Rightarrow Q_{2n}(z) \neq 0 \\ z = 0 : Q_{2n}(z) &= z^n P(z + z^{-1}) \\ &= z^n [a_n (z + z^{-1})^n + a_{n-1} (z + z^{-1})^{n-1} + \dots] \\ \therefore Q_{2n}(0) &= a_n \end{aligned}$$

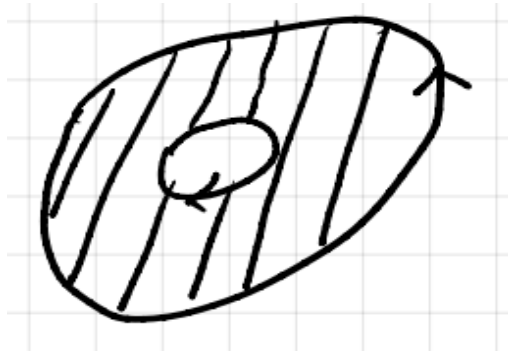
$$\Rightarrow Q_{2n}(z) \neq 0 \text{ for any } z$$

So the function is analytic, therefore

$$\int_0^{2\pi} \frac{d\theta}{p(2\cos\theta)} = 0$$

Which is a contradiction

□



For multiply connected regions,

$$C = C_0 \cup C_1 \cup C_2 \cup C_3$$

$$\begin{aligned} \oint_C f(z) dz = 0 &= \underbrace{\oint_{C_0} f(z) dz}_{\text{counter clockwise}} + \underbrace{\oint_{C_1} f(z) dz}_{\text{clockwise}} + \underbrace{\oint_{C_2} f(z) dz}_{\text{clockwise}} + \underbrace{\oint_{C_3} f(z) dz}_{\text{clockwise}} \\ &= \oint_{C_0} f(z) dz - \oint_{C_1} f(z) dz - \oint_{C_2} f(z) dz - \oint_{C_3} f(z) dz \end{aligned}$$

The outside contour C_0 is counter clockwise, The inside contours C_1 , C_2 , C_3 have clockwise direction.